engineering 4 us all

Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element E Initial Template - AT - 15 points

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements *(E1)*

High-Priority Design Requirement #1 from Element C - Durable and Portable

STEM Concepts Involved in the requirement: Engineering & Material Science: Choosing strong but lightweight materials.
Physics: Understanding how weight and force affect the structure.
Math: Measuring weight and balance to keep the system stable.

Describe how STEM concepts apply directly to the specifics of your design requirement:
We need our plant watering system to be strong enough to last a long time but light enough to move if needed. We're using PVC pipes because they are durable and won't rot or rust. The plastic tubing is flexible and lightweight, so it won't add too much weight. The system needs to be stable so it doesn't tip over or break. To make sure it's balanced, we consider how weight is spread out and make sure the base is strong.

High-Priority Design Requirement #2 from Element C - Reliable Water Supply

Name: ______________________________ **Date:** _________________ **Class:** _______________

STEM Concepts Involved in the requirement:Biology: How much water do plants need to grow properly?
Physics (Fluid Mechanics): How water moves through tubes using gravity and pressure.
Math: Measuring the right amount of water to prevent overwatering or underwatering.

Describe how STEM concepts apply directly to the specifics of your design requirement:
Plants need a steady water supply, so we designed a system that lets water drip slowly into the soil. We use IV needles and plastic tubing to control how much water flows out. Gravity helps move the water from a small tank to the plants. If the water flows too fast, it could flood the plants, and if it's too slow, they might dry out. By testing the flow rate and adjusting the tubing, we make sure the plants get just the right amount of water.

High-Priority Design Requirement #3 from Element C - Nutrient-Rich Soil

STEM Concepts Involved in the requirement: Chemistry: How different nutrients help plants grow.
Biology: How plants absorb nutrients and use them for photosynthesis.
Math: Keeping track of soil pH levels and fertilizer amounts.

Describe how STEM concepts apply directly to the specifics of your design requirement:
Plants need nutrients in the soil to grow healthy. If the soil doesn't have the right balance of nutrients, plants can become weak or die. We will check the pH level to make sure it's in the right range (between 5 and 7). If the soil is too acidic or too basic, we can add lime or sulfur to fix it. We might also use compost or fertilizer to keep the soil rich in nutrients. This helps plants develop strong roots and healthy leaves.

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High-Priority Design Requirement #4 from Element C (if you have one)-Realistic Size
STEM Concepts Involved in the requirement:Engineering: Making sure the system fits in the classroom.
Math: Measuring space to make sure everything fits properly.
Physics: Keeping the system stable so it doesn't fall over.

Describe how STEM concepts apply directly to the specifics of your design requirement:
Our plant system needs to be big enough to hold the plants and water but small enough to fit in the classroom. We use math to measure the space and make sure the system isn't too large or heavy. The base needs to be stable, so we make sure the weight is spread out evenly so it won't tip over when we add water.

Proposed Solution to STEM Principles and Practices *(E2)*

STEM Principles for Design Solution #
STEM Concept:*Water Flow and Gravity*

Describe how STEM concept applies directly to the specifics of the first solution to your problem: Our system uses gravity to move water from the source to the plants. The IV tubing controls the flow rate, allowing water to drip slowly instead of pouring all at once. This prevents overwatering and ensures a steady water supply. Fluid mechanics helps us understand how water pressure changes based on tube size and height differences. By adjusting the height of the water source, we control how fast water moves.

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You can only work on the next sections after you have built and tested your first prototype. You need to skip to the “Evidence is Reviewed” section.

STEM Principles for Design Solution #2 STEM Concept: Material Strength and Stability Describe how STEM concept applies directly to the specifics of your solution: Our design uses PVC pipes and plastic tubing because they are lightweight, strong, and water-resistant. This keeps the system durable and easy to move. We also make sure the base is wide and stable so the system doesn’t tip over when water is added. Forces and balance in physics help us understand how to distribute weight evenly to keep the structure upright.

Evidence Design is Reviewed (E3)

Documentation #1 (Only one documentation section needed for full credit):

What format: Interview notes / Mr. Butler

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- When explaining the length of time and durability, make sure that the time length isn't too broad. Make sure that when you set a time length, there is a goal and specific time, whether it be 3-5 years, make sure there is a specification and detail.

Documentation #2 (+ making only):

What format: Interview notes, emails, audio/video tape with transcript, etc.

- What the expert said about the connection you've made from your design requirements to STEM principles and practices:
- What the expert said about how STEM principles and practices substantiate your design solution:

Documentation #3 (+ making only):

What format: Interview notes, emails, audio/video tape with transcript, etc.

- What the expert said about the connection you've made from your design requirements to STEM principles and practices:
- What the expert said about how STEM principles and practices substantiate your design solution:

Experts with Credentials *(E4)*

Document who your field experts are (names are not necessary) and what qualifies them as an expert. Use as many copies of this sub-template as needed.

Expert #1 (from documentation 1): The expert is a professional engineer with experience in designing and implementing structural and mechanical solutions.

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Expert #2 (from documentation 2): Describe credential(s) and what makes them qualified as an expert.

Expert #3 (from documentation 3): Describe credential(s) and what makes them qualified as an expert.

Verification or Corrective Response to Design Review *(E5)*

Your conversation/interview with the expert(s) should inform some corrective responses. In E5, describe how those reviews informed your responses.

For example,

"_________ told us _________, which informed our decision to change/reconsider _________ ."
(or "...which verified/reinforced our decision to _________.")

Make sure to document the experts' reviews (suggestions/feedback or criticisms of your proposed solution) and how they will inform iterations you will make to your proposed solution. Use as many copies of this sub-template as needed.

Interview with Expert #1 informs corrective response:

Interview with Expert #2 informs corrective response:

Interview with Expert #3 informs corrective response:

	Criteria	*5*	*4*	*3*	*2*	*1*	*0*
	3. Evidence that Design is Reviewed. There is ______________ evidence that the	*substantial*	*adequate*	*some*	*minimal*	*no - but review was requested -*	*no*

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	application of STEM principles & practices (or suitable alternative) has been reviewed by expert(s).						
	4.**Design Reviewed by Experts.** _____________ expert(s) have/has reviewed the design for application of STEM principles & practices.	*Two or more*, with *at least two* with documented, relevant credentials	*Two or more*, with *at least one* with *documented, relevant* credentials	*At least one* with *documented, clearly relevant* credentials	*At least one* with *documented* credentials	*Only one with unclear credentials*	*No*
	5.Design Review Results in Verification. The design reviews provided _____________.	*clear and detailed* confirmation (verification) or *substantial detail* necessary to inform a corrective response	*clear* confirmation (verification) or *detail* necessary to inform a corrective response	*partial* confirmation (verification) or *some detail* necessary to inform a corrective response	*does not provide clear* confirmation (verification) or suggestions for a corrective response, *but at least some response is provided*	show no evidence of confirmation (verification) or suggestions for a corrective response by an expert, *but includes notes of attempts to gain feedback*	*show no evidence of confirmation (verification) or suggestions for a corrective response by an expert*